Notes: Henderson (AER, 1996)
Effects of Air Quality Regulation

Contents

1 Big picture

- **Research question.** How does local implementation of U.S. ground-level ozone regulation affect air quality, industrial location, and the timing of economic activity?
- **Core institutional setting.** Counties are designated as in attainment or nonattainment of national ambient air quality standards. Nonattainment status triggers stricter local regulatory effort, monitoring, and restrictions on firms.
- **Main empirical idea.** Cross-sectional comparisons are misleading because polluted places are exactly the places with regulation and polluting industries. Henderson uses panel variation in attainment status, fixed effects, and changes over time.
- **Main findings.** Switching to nonattainment status induces greater regulatory effort and improves ozone readings by roughly 3–8 percent, depending on the air-quality measure. Pro-environment states also have cleaner air.
- **Industrial response.** Polluting industries tend to move from regulated, dirty nonattainment counties toward cleaner attainment counties. This can reduce local regulatory scrutiny without necessarily reducing total emissions.
- **Behavioral response to the standard.** Because the ozone standard is based on an extreme hourly reading, localities can satisfy the letter of the regulation by dampening daily peaks even when typical ozone conditions, such as mean or median readings, do not improve.
- **Economic takeaway.** Environmental regulation matters, but the exact design of the standard matters as well. A standard based on a peak reading creates incentives to manage peaks, relocate activity, and shift activity over the day.

2 Incremental contribution of the paper

- **From cross-section to panel evidence.** Earlier work often found weak or perverse effects of air-quality regulation because regulation, pollution, and polluting industries are positively correlated across places. Henderson uses within-location changes to separate regulatory effects from permanent local pollution differences.
- **Three outcomes rather than one.** The paper studies air quality, firm location, and within-day timing of activity. This makes it broader than a study of only pollution levels or only plant location.
- **Direct link to regulatory design.** The paper shows that a rule based on the second-highest daily maximum can improve that extreme-value measure while leaving mean hourly ozone essentially unimproved.
- **Local implementation matters.** The paper emphasizes that national standards operate through local and state-level enforcement. County attainment status and state regulatory attitudes both affect measured outcomes.

- **Relocation channel.** The paper documents that high-emitting industries expand more in counties with a clean attainment record, consistent with firms avoiding tougher regulation.
- **Policy lesson.** Environmental standards should be evaluated not only by whether the official compliance statistic improves, but also by how agents reallocate activity in response to the rule.

3 Data and measurement

3.1 Air-quality data

The paper uses hourly ozone readings from U.S. Environmental Protection Agency monitoring stations for 1977, 1982, 1985, and 1987. The main panel contains 643 monitoring stations that report in at least two of the four years and are located in urban counties.

- **Pollutant.** Ground-level ozone, denoted O_3, which is formed from volatile organic compounds, nitrogen oxides, and atmospheric conditions such as temperature.
- **Regulatory measure.** The official ozone standard is based on the annual second-highest daily maximum hourly reading. A county is in attainment if the highest hourly reading exceeds 0.12 parts per million on no more than one day per year.
- **Additional measures.** Henderson also studies the annual mean hourly reading, the median of daily maximum readings in July, and the mean July hourly reading.
- **Why July?** Annual measures are affected by differences in the number of days monitored. July has high temperatures and nearly universal hourly readings, so July measures reduce concerns about changes in monitoring coverage.

3.2 Regulatory variables

- **Ozone nonattainment status.** A county-level indicator for whether the county is in ozone nonattainment, often lagged one year to allow regulatory effort to take effect.
- **Other-pollutant nonattainment index.** An index measuring the extent of nonattainment in sulfur oxides, carbon monoxide, nitrogen oxides, and total suspended particulates.
- **State regulatory attitude.** A state-level fixed effect from a model of pollution-abatement expenditures. States that spend more than predicted on abatement are interpreted as more pro-environment and more enforcement-oriented.

3.3 Economic and geographic controls

- **Weather.** Average daily maximum July temperature is a key control, because ozone formation is temperature sensitive.

- **Employment.** Metropolitan employment controls for general economic activity, commuting, and emissions-generating activity.
- **Industrial composition.** The presence of major volatile organic compound emitters is included, such as plastics, industrial organic chemicals, and petroleum refining.
- **Geography.** County land area, coastal location, and a Los Angeles coastal corridor dummy capture dispersion, wind patterns, and special regional pollution conditions.

3.4 Industrial-location data

For the relocation analysis, the paper studies five three-digit SIC industries that are major volatile organic compound emitters:

- plastic materials and synthetics;
- industrial organic chemicals;
- miscellaneous plastics;
- petroleum refining;
- blast furnace and primary steel.

The spatial unit is the urban county. The paper studies how the number of establishments in these industries changes across counties with different attainment histories.

4 Methodology and empirical specifications

4.1 Station fixed-effects air-quality model

The main air-quality model is

$$O3_{it} = C + \beta X_{it} + \gamma Z_i + f_i + \varepsilon_{it},$$

where $O3_{it}$ is an annual ozone summary measure for monitoring station i in year t, X_{it} contains time-varying controls and regulatory variables, Z_i contains time-invariant geographic and state-attitude variables, and f_i is a station fixed effect.

- The fixed effects absorb permanent differences in monitor location, local geography, and baseline pollution conditions.
- The coefficient on nonattainment status is identified from counties that change attainment status over time.
- Hausman tests reject random effects, so fixed effects are essential.

4.2 Recovering time-invariant effects

Because fixed effects remove time-invariant variables, Henderson recovers their effects in a second step. After estimating $\hat{\beta}$, he constructs the station-level average residual

$$\overline{RES}_i = \overline{O3}_i - \hat{\beta}\overline{X}_i = C + \gamma Z_i + f_i + \frac{1}{T}\sum_t \varepsilon_{it}.$$

He then regresses $\overline{RES}_i$ on the time-invariant variables Z_i, including state regulatory attitude and geography.

4.3 Industrial-location model

For the plant-location analysis, the paper estimates a reduced-form model for the number of establishments in industry s, county j, and year t. The dependent variable is the natural log of the number of establishments, with special handling of counties with zero plants.

- The key explanatory variable is a dummy equal to one if the county has been in attainment for the last three years.
- The central hypothesis is that a clean attainment record attracts polluting plants by reducing regulatory scrutiny.
- Henderson estimates fixed-effects Tobit models and checks robustness using conditional fixed-effects Poisson models.

4.4 Hourly ozone model

To study within-day timing, the paper estimates pooled hourly models for July readings in 1982 and 1987:

$$O3_{st} = A + \sum_{i=1}^{m} \alpha_i O3_{s,t-i} + \sum_{j=1}^{23} D_{jt}\beta_j + A_{87}D_{87} + \sum_{j=1}^{23} D_{jt}\gamma_j + \cdots + \varepsilon_{st}.$$

- Lagged ozone terms capture persistence and accumulation of ozone over the day.
- Hour dummies capture the daily cycle of emissions-generating economic activity.
- 1987 differential hourly dummies show how the within-day activity cycle changed from 1982 to 1987.
- The model is estimated separately for nonattainment stations, attainment stations, and California stations.

5 Identification and interpretation

5.1 What the paper can identify

The strongest identification comes from changes within monitoring stations and counties over time. A county switching from attainment to nonattainment status is compared to itself before and after the switch, while controlling for weather, employment, industrial composition, and time effects.

5.2 Why cross-sections are misleading

Dirty areas are more likely to be designated nonattainment and more likely to contain polluting industries. A pure cross-section could therefore incorrectly suggest that regulation causes pollution or attracts polluters. The panel design is meant to separate this baseline association from changes induced by regulation.

5.3 Endogeneity concerns

Nonattainment status is not randomly assigned. It is triggered by pollution readings. Henderson addresses this by using fixed effects, lags of regulatory status, controls for local economic conditions and weather, and tests of strict exogeneity. These checks strengthen the interpretation but do not make the design a randomized experiment.

5.4 Measurement issue: days monitored

Annual ozone means and maxima depend on how many days a station records readings. Because monitoring coverage can itself respond to pollution conditions, days recorded are difficult to treat as exogenous. The July measures help address this issue because coverage is nearly universal in July.

5.5 Interpretation of the nonattainment coefficient

The nonattainment coefficient should be interpreted as the effect of a regulatory regime, not a single rule. Nonattainment triggers a bundle of stricter permitting, monitoring, equipment requirements, emissions reductions, and possibly transportation-related rules.

5.6 Welfare interpretation

The industrial relocation result is not necessarily welfare-improving. Moving polluting activity from dirty nonattainment counties to cleaner attainment counties can narrow the distribution of peak ozone readings, but it may spread emissions rather than reduce them. This is one reason the paper emphasizes incentives created by the exact design of the standard.

6 Tables and figures

6.1 Figures 1 and 2: Extreme-value ozone versus mean ozone

Read Figure 1:

- Figure 1 plots the distribution of annual second-highest daily maximum ozone readings for 1977, 1982, and 1987.
- The distribution shifts left and narrows after the 1977 Clean Air Act amendments.
- The peak of the distribution moves near or below the 0.12 parts-per-million standard.

Read Figure 2:

- Figure 2 plots the distribution of annual mean hourly ozone readings.
- Unlike the extreme-value measure, the mean hourly reading does not clearly improve.
- The figure illustrates the paper's main regulatory-design point: the compliance measure can improve even when typical conditions do not.

Economic message: Regulation successfully targets the official extreme-value statistic, but that does not necessarily imply broad improvement in typical ozone exposure.

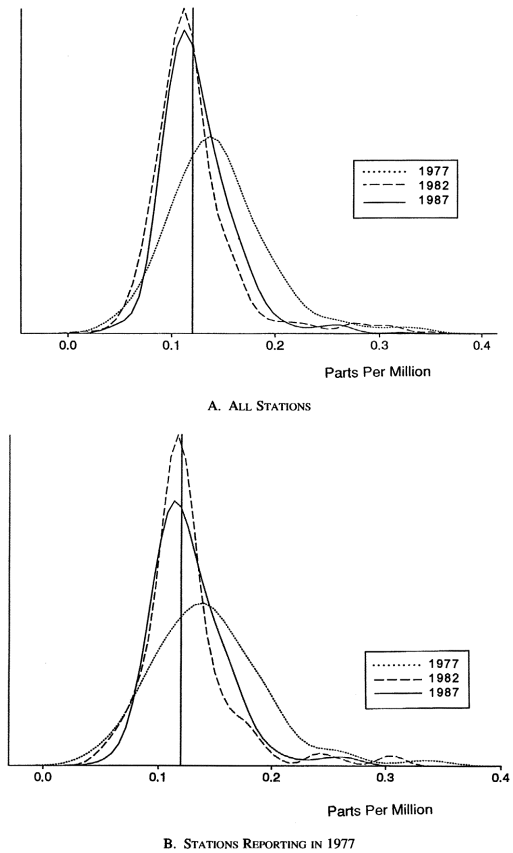


Figure 1. Distribution of Annual Second-Highest Daily Maximum

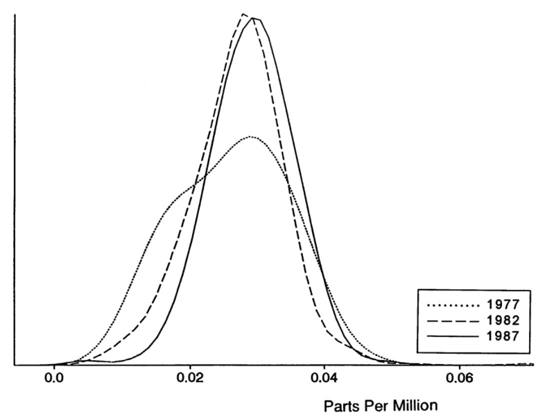


Figure 2. Distribution of Annual Mean of Hourly Readings

Figure 1: Distribution of annual second-highest daily maximum ozone readings.

Figure 2: Distribution of annual mean hourly ozone readings.

6.2 Tables 1 and 2: Air-quality determinants

Read Table 1:

- Table 1 reports station fixed-effects estimates for four ozone measures.
- July temperature strongly raises ozone readings.
- The presence of industrial organic chemicals raises ozone readings across all measures.
- Ozone nonattainment status has negative coefficients for all measures and is statistically significant for the median of daily July maxima.
- Nonattainment in other air-quality dimensions is significantly associated with lower annual ozone measures, consistent with broader regulatory pressure.

Read Table 2:

- Table 2 studies time-invariant geography and state regulatory attitude.

- States with stronger pro-environment regulatory attitudes have cleaner annual mean, July median, and July mean readings.
- The state-attitude effect is not significant for the second-highest daily maximum, which may reflect direct targeting of the official compliance statistic even by weaker-enforcement states.
- The Los Angeles coastal dummy is strongly positive, reflecting the special severity of the Los Angeles ozone problem.

Economic message: Local regulation matters, but enforcement intensity and geography also matter. Nonattainment status and state attitudes capture different layers of the regulatory system.

TABLE 1—TIME-VARIANT DETERMINANTS OF AIR QUALITY

	ln(second highest daily maximum)	ln(mean annual reading)	ln(median of daily maximum July)	ln(mean July reading)
ln(average daily maximum temperature July)	1.09*	0.213	1.85*	1.20*
	(0.179)	(0.164)	(0.190)	(0.187)
ln(metropolitan employment)	0.080	0.090	0.137[a]	0.212*
	(0.067)	(0.062)	(0.073)	(0.072)
Presence of plastics	0.109*	0.032	−0.070	−0.055
	(0.056)	(0.051)	(0.060)	(0.058)
Presence of organic industrial chemicals	0.046*	0.058*	0.063*	0.092*
	(0.022)	(0.020)	(0.023)	(0.023)
Presence of petroleum refining	0.019	0.014	−0.0061	−0.0076
	(0.018)	(0.017)	(0.019)	(0.019)
Year 1977	0.171*	0.014	0.133*	0.115*
	(0.020)	(0.018)	(0.022)	(0.022)
Year 1985	−0.013	0.041*	−0.0037	0.027[a]
	(0.013)	(0.012)	(0.014)	(0.014)
Year 1987	0.012	0.058*	−0.036*	−0.00014
	(0.016)	(0.015)	(0.017)	(0.017)
County in ozone	−0.026 −0.017	−0.029 −0.020	−0.078* −0.081*	−0.043[a] −0.040[a]

Table 1: Time-variant determinants of air quality.

TABLE 2—THE IMPACT OF TIME-INVARIANT VARIABLES

	Second-highest daily reading	Annual mean	Median of July daily maxima's	July mean
Constant	−1.33*	1.27*	−5.64*	−4.33*
	(0.020)	(0.022)	(0.028)	(0.031)
Dummy for county on coast	0.083*	−0.164*	−0.140*	−0.274*
	(0.018)	(0.022)	(0.027)	(0.032)
Dummy for Los Angeles coast	0.454*	0.114*	0.519*	0.295*
	(0.043)	(0.055)	(0.061)	(0.065)
County land[a] Area	-0.314×10^{-4}*	-0.329×10^{-4}*	0.598×10^{-4}*	-0.486×10^{-4}*
	(9.99×10^{-6})	(0.124×10^{-4})	(0.131×10^{-4})	(0.169×10^{-4})
County land Area squared	2.35×10^{-9}*	1.67×10^{-9}*	4.01×10^{-9}*	3.19×10^{-9}*
	(6.43×10^{-10})	(7.08×10^{-10})	(7.59×10^{-10})	(8.51×10^{-10})
State attitude (state fixed effects)	0.019	−0.042*	−0.038*	−0.052*
	(0.013)	(0.014)	(0.019)	(0.020)
Adjusted R^2	0.291	0.097	0.157	0.139
N	643	643	616	616

Note: Standard errors are in parentheses.

Table 2: Time-invariant determinants of air quality.

6.3 Tables 3 and 4: Industrial relocation

Read Table 3:

- In 1978, most plants in the five high-emitting industries were located in nonattainment counties.
- From 1978 to 1987, plant growth in clean attainment counties exceeds plant growth in counties that remain nonattainment.
- This contrast is especially striking because overall manufacturing-establishment growth and employment growth are similar across the county groups.

Read Table 4:

- Table 4 estimates the effect of a clean three-year attainment history on the number of establishments in polluting industries.
- Clean counties have significantly more establishments in industrial organic chemicals, petroleum refining, plastic materials, and miscellaneous plastics.
- Steel is not significantly affected, consistent with steel being a smaller volatile organic compound emitter than the other industries.

- For plastics, nonattainment in other air-quality dimensions is associated with large reductions in establishments.

Economic message: Polluting plants do not just clean up in place. They tend to reallocate toward counties with a cleaner regulatory record, which is outside the main intent of the Clean Air Act.

TABLE 3

A. 1978 Stock of Plants

Industry	Standard industrial classification (SIC)	Percent of plants located in nonattainment counties
Plastic materials and synthetics	282	91
Industrial organic chemicals	286	89
Miscellaneous plastics	307	87
Steel	331	92
Petroleum refining	291	92
Percent of all urban counties with 1978 nonattainment status = 60 percent		

B. Total Percentage Change in Number of Plants 1978–1987 in Counties by Ozone Attainment Status

Industry (SIC)	Attainment in both 1978 and 1987	Nonattainment in both 1978 and 1987	Nonattainment in 1978 attainment in 1987
Plastic materials and synthetics (282)	67	14	21
Industrial organic chemicals (286)	19	8	36
Miscellaneous plastics (307)	69	20	38
Primary steel (331)	28	2	6
Petroleum refining (291)	15	6	−5

Table 3: Initial location and growth of high-polluting plants.

TABLE 4—INDUSTRIAL LOCATION: THE NATURAL LOG OF THE NUMBER OF ESTABLISHMENTS IN COUNTY j IN YEAR t

	Industrial organic (286)	Petroleum refining (291)	Plastic materials (282)		Miscellaneous plastics (307)		Steel (331)
Dummy:	(i)	(i)	(i)	(ii)	(i)	(ii)	(i)
Clean for last three years	0.091*	0.065*	0.072*	0.056[a]	0.081*	0.065*	−0.003
	(0.029)	(0.038)	(0.031)	(0.033)	(0.019)	(0.019)	(0.026)
Index of nonattainment in other air quality dimensions				−0.105*		−0.115*	
				(0.038)		(0.027)	
ln(metropolitan area employment all other industries)	0.046	1.02*	0.159[a]		0.326*		0.112
	(0.072)	(0.079)	(0.082)		(0.092)		(0.071)
N [nonzero, (zero)]	1991 (609)	1179 (725)	1869 (796)		4903 (521)		2462 (802)

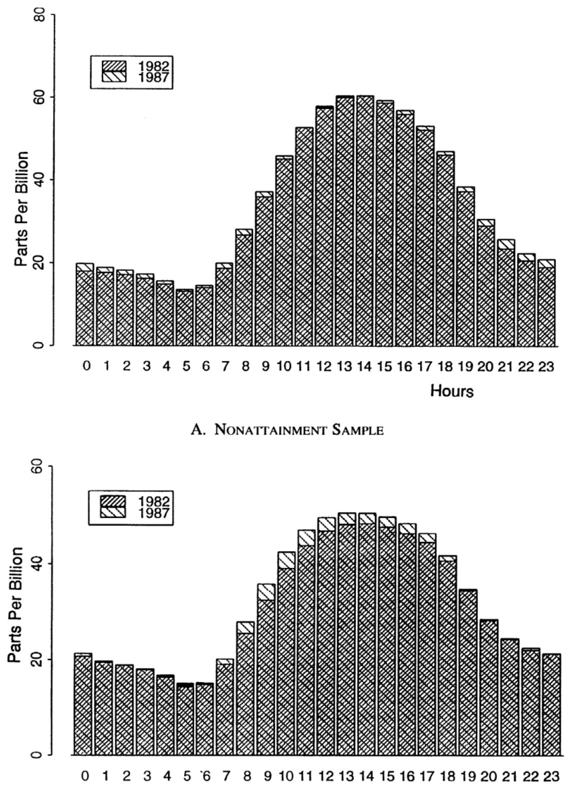
Table 4: Industrial location and clean attainment history.

6.4 Figure 3: Average hourly ozone readings

Read:

- Figure 3 shows average hourly July ozone readings for nonattainment, attainment, and California samples.
- Ozone readings follow a strong daily cycle and tend to peak around midday or early afternoon.
- The California sample is especially informative because it contains some of the most severe ozone problems and shows strong evidence of within-day rescheduling.

Economic message: Daily ozone exposure is shaped by the interaction between the timing of economic activity and the atmospheric lag through which emissions become ozone.

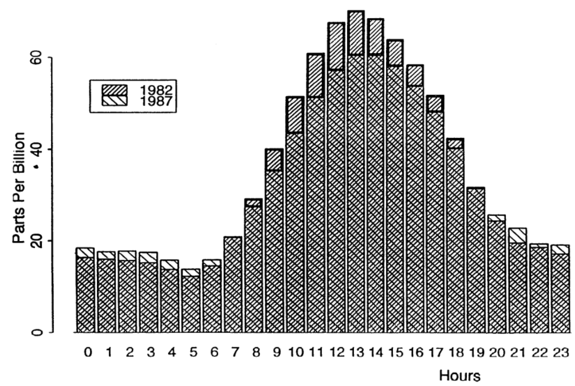


A. Nonattainment Sample

B. Attainment Sample

Figure 3. Average of Hourly Ozone Readings (July)

Figure 3, panels A–B: Nonattainment and attainment samples.



C. California Sample

Figure 3. *Continued.*

Figure 3, panel C: California sample.

6.5 Figure 4: Daily ozone pattern

Read:

- Figure 4 plots observed ozone, predicted ozone, and estimated hourly shocks for the 1982 nonattainment sample.
- The estimated economic-activity shock peaks around 8 a.m.
- Ozone peaks later, around 1–2 p.m., because ozone accumulates with a lag.

Economic message: Reducing morning emissions-generating activity can reduce later ozone peaks even if total daily activity is not reduced.

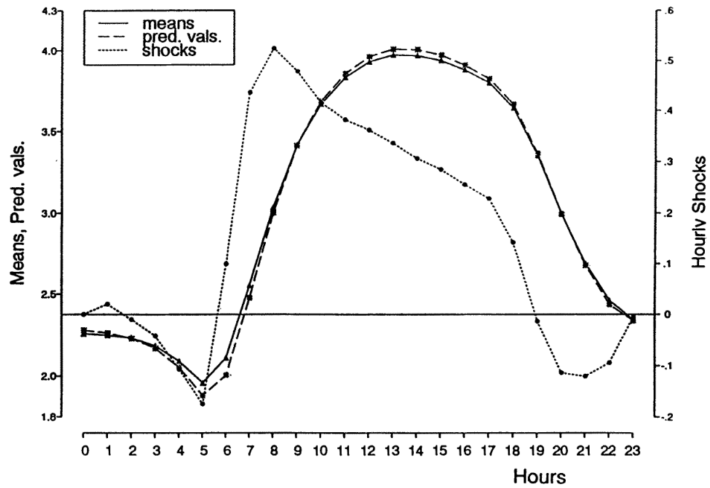


Figure 4: Daily ozone pattern in the 1982 nonattainment sample.

6.6 Figure 5 and Table B1: Hourly shocks and ozone persistence

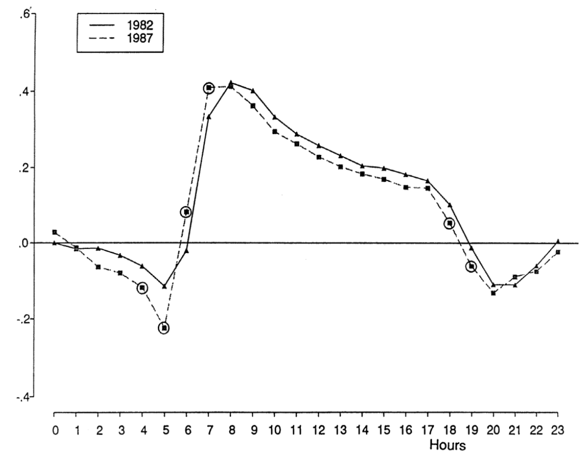
Read Figure 5:

- Figure 5 compares estimated hourly shocks in 1982 and 1987.
- Nonattainment areas show reductions in key morning shocks from roughly 8–10 a.m. and increases in some other hours.
- California shows a particularly clear shift away from the critical 7–10 a.m. window.
- These changes help explain how peak ozone readings can be dampened without reducing mean hourly ozone.

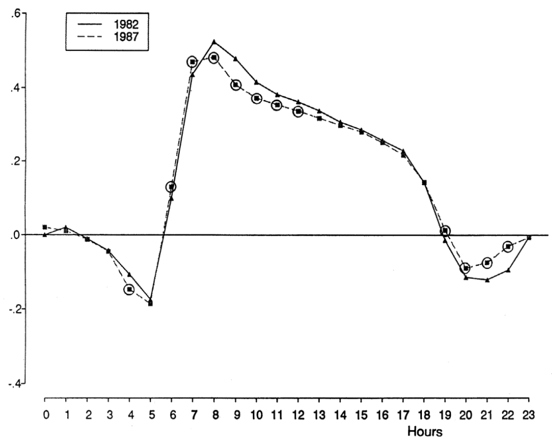
Read Table B1:

- Table B1 reports coefficients on lagged hourly ozone readings.
- The previous hour's ozone reading is highly persistent, with coefficients between about 0.83 and 0.95 across samples.
- Because ozone is persistent, morning shocks have important indirect effects on later ozone levels.

Economic message: The local response is consistent with strategic timing. By shifting emissions away from hours that generate peak ozone, localities can comply with an extreme-value rule while leaving average conditions less improved.

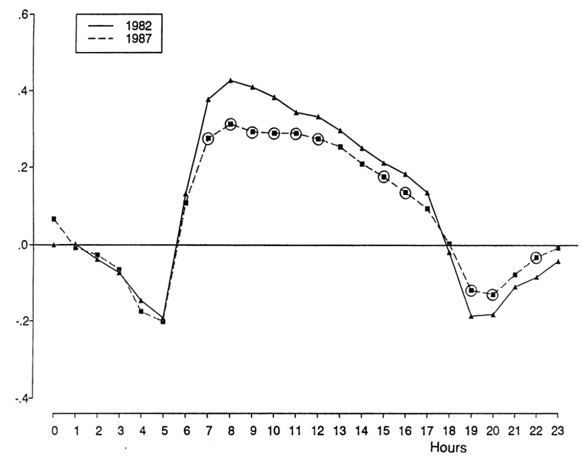


B. ATTAINMENT SAMPLE



A. NONATTAINMENT SAMPLE

FIGURE 5. HOURLY SHOCKS TO OZONE



C. CALIFORNIA SAMPLE

FIGURE 5. *Continued.*

Figure 5, panel A: Hourly shocks in the nonattainment sample.

Figure 5, panels B–C: Hourly shocks in attainment and California samples.

TABLE B1—COEFFICIENTS FOR LAGGED DEPENDENT VARIABLES

Lagged dependent variables	Nonattainment areas	Attainment areas	California
α_1	0.9467*	0.9323*	0.8301*
	(0.0015)	(0.0035)	(0.0032)
α_2	−0.0979*	−0.0804*	0.0413*
	(0.0021)	(0.0048)	(0.0042)
α_3	0.0044*	0.0074	−0.0147*
	(0.0021)	(0.0048)	(0.0042)
α_4	−0.0280*	−0.0030	−0.0613*
	(0.0015)	(0.0036)	(0.0032)

Note: Standard errors are given in parentheses.
 * Significant at the 5-percent level.

Table B1: Coefficients for lagged dependent variables.

7 Short summary

- The paper studies U.S. ground-level ozone regulation from 1977 to 1987.
- The key regulatory variable is county attainment or nonattainment status under the ozone standard.
- Nonattainment status triggers stricter local regulatory effort and is associated with cleaner ozone readings after controlling for station fixed effects, weather, employment, and industry composition.
- Pro-environment state regulatory attitudes are associated with cleaner air.
- High-emitting industries tend to relocate toward counties with a clean attainment record.
- The official ozone standard is based on an extreme hourly reading, not mean exposure.
- Extreme-value readings improved substantially, but mean hourly ozone readings did not improve in the same way.
- Localities appear to reduce peak readings partly by shifting activity away from the morning hours that generate later ozone peaks.
- The paper's policy lesson is that the design of regulation shapes behavioral responses, not just compliance outcomes.

8 Conclusion

8.1 Core conclusion

Henderson shows that local air-quality regulation matters. Counties that enter ozone nonattainment face greater regulatory pressure and subsequently experience cleaner measured ozone readings. States with stronger regulatory attitudes also have cleaner air.

8.2 Economic intuition

Regulation changes incentives. Firms in dirty, regulated areas face higher compliance costs, monitoring, and permitting burdens, so polluting industries have incentives to shift toward counties with a cleaner attainment record. Localities also respond to the exact form of the ozone standard: when compliance depends on a single extreme hourly reading, reducing the relevant peak can be more attractive than reducing total emissions.

8.3 Incremental contribution revisited

The paper's contribution is to show that environmental regulation has real effects on both air quality and economic geography, while also demonstrating that agents respond strategically to the definition of the regulated metric. This links environmental economics, industrial location, and policy design.

8.4 Implications

The findings imply that regulators should evaluate environmental policy using multiple air-quality statistics, not just the compliance trigger. They should also account for spillovers across space and time: firms may move across counties, and emissions-generating activity may move across hours of the day.

8.5 Limitations

The empirical design is observational and based on regulatory status changes, not random assignment. Nonattainment status is tied to pollution history. The paper uses fixed effects, lags, controls, and robustness checks to address this, but some endogeneity concerns remain. In addition, the nonattainment variable captures a bundle of regulatory actions rather than a single policy instrument.

8.6 Takeaway

The enduring takeaway is that regulation works, but it works through incentives. Ozone regulation improved the official peak-based air-quality measure, pushed polluting industries toward cleaner counties, and encouraged within-day rescheduling that reduced peak readings without necessarily improving average ozone conditions.